Exercise 1 SQL Fundamentals

1) SELECT *
FROM employees

*NB THE OUTCOME is the whole given table

2) SELECT DISTINCT
FROM employees

Department
IT
HR
Finance
Marketing

3) SELECT first_name,
last_name,
Salary
FROM employees
ORDER BY Salary DESC;

First_name	last_name	Salary
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brown	53 000
Daniel	Clark	53000
Daniel	White	52 000
Jessica	Moore	51000
Chris	Hall	50000
Jane	Smith	48000

4 SELECT*
FROM employees
ORDER BY Salary DESC,
LIMIT 5;

Id	First_name	last_name	Department	Salary	Hire Date	City
6	Emily	Davis	IT	62000	14-02-15	Chicago
3	Mike	Johnson	Finance	60 000	30-09-2017	Los Angeles
7	Robert	Wilson	Finance	59000	09-10-2019	Houston
1	John	Doe	IT	55000	15-06-2018	New York
4	Sarah	Brown	IT	53000	25-03-2021	New York

5 SELECT*
FROM employees
WHERE department = IT;

Id	First_name	last_name	Department	Salary	Hire-date	City.
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

6 SELECT*
FROM employees
WHERE Department = finance,
AND salary > 58000;

Id	first_name	last_name	Department	Salary	Hire date	City
3	Mike	Jackson	finance	60 000	2017-09-30	Los Angeles
7	Robert	Wilson	finance	59 000	2019-10-09	houston

7 SELECT *
FROM employees
WHERE department = HR
OR Department = Marketing ;

Id	first_name	last_name	Department	Salary	Hire date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-0	Chicago

8 SELECT *
FROM employees
WHERE Department != IT ;

Id	First_name	last_name	Department	Salary	Hire-date	City
2	Jane	Smith	HR	48000	2017-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
5	David	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	59000	2018-05-22	Los Angeles
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-29	Chicago

9 SELECT *
FROM employees
WHERE Department = HR
OR Department = IT
OR Department = Finance;

Id	First_name	Last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50000	2020-08-10	San-Francisco

16 SELECT *
FROM employees
WHERE Department = IT
AND Salary > 50 000
AND City = "new york";

Id	First_name	Last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

11

```
SELECT   first_name,
         last_name
FROM     employees
WHERE    Department = Finance
  OR     Department = Marketing
  AND    Salary > 52000
ORDER BY Salary DESC;
```

first_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

12

```
SELECT   DISTINCT City
FROM     employees
WHERE    Department != IT
  AND    Department != HR;
```

City
Chicago
Los Angeles
San Francisco
Houston

13 SELECT *
FROM employees
WHERE Department ! = Finance
AND Salary > 50000
ORDER BY hire-table ASC;

Id	first-name	last-name	Department	Salary	Hire-date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

14 SELECT *
from employees
WHERE City = "Chicago"
OR City = "Los Angeles"
AND Department = "IT"
OR Department = 'Marketing'
LIMIT 3;

Id	First_name	last_name	Department	Salary	Hiredate	City
~~2~~	~~Jane~~	~~Smith~~	~~H~~			
6	Emily	Davis	IT	62000	15-02-14	Chicago
9	Daniel	Clark	Marketing	53000	22-06-0	Chicago